Summary

1. Enzymes increase the rate of a reaction by lowering the energy of activation.
2. The active site of an enzyme is a pocket or cleft containing aminoacid side chains that participate in the binding of substrate and the catalytic reaction.
3. Binding of substrate causes a conformational change in the enzyme (induced fit) that favours catalysis.
4. When a substrate binds to the active site, an enzyme–substrate (ES) complex is formed which subsequently dissociates to enzyme and product.
5. Most enzymes show Michaelis-Menten kinetics, and a plot of the initial reaction velocity (v) against substrate concentration([S] has a hyperbolic graph.
6. Michaelis Menten derived the equation for a single substrate enzyme catalyzed reaction assuming steady state conditions where there is no change in [ES].
7. Lineweaver Burk used the double reciprocal linear plot to determine V_{max} and K_m more accurately.
8. Inhibitors are molecules that can reduce the rate of an enzyme catalyzed reaction. There are two types of inhibitors; reversible and irreversible inhibitors.
9. Competitive (V_{max} unchanged, K_m increased), non-competitive (V_{max} decreased, K_m unchanged) and uncompetitive inhibitors (Both V_{max} and K_m decreased) are reversible inhibitors.
10. Allosteric enzymes do not show Michaelis Menten kinetics, instead they show sigmoidal kinetics. Allosteric inhibitors bind to a site other than the active site.
11. Irreversible inhibitors bind tightly to the enzyme, either covalently or noncovalently, and dissociates very slowly from the enzyme - inhibitor complex.

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